Xi'an Jiaotong-Liverpool University
西交利物浦大学

Department of Biological Sciences

BIO204 Advanced Genetics - Lab Report 2

Name of Practical: Bacterial Mutagenesis

Group Number: 35 (Thursday)

	Name	XJTLU ID Number
Student 1	Yuewei Xu	
Student 2	Yanbing Yang	
Student 3	Muge Zhang	

	Student 1	Student 2	Student 3
Peer Mark by Student 1		100%	100%
Peer Mark by Student 2	100%		100%
Peer Mark by Student 3	100%	100%	
Relative Peer Contribution	100%	100%	100%
Group Lab Report Mark			
Individual Lab Report Mark			

WP2 trp- Reverse Mutation Assay Reveals Distinct Spontaneous, MMS-Induced and UV-Induced Mutagenic Patterns in *Escherichia coli* – Balance Between Mutagenicity and Cytotoxicity

1. Abstract

The WP2 trp- tryptophan reverse mutation assay was used to quantify spontaneous mutation frequency and assess mutagenesis induced by methyl methanesulphonate (MMS) and ultraviolet (UV) radiation in *Escherichia coli*. Viable cell concentration, estimated from nutrient agar counts, was 2.6×10^{8} CFU/ml. The true spontaneous mutation frequency, determined on SA3 plates lacking tryptophan, was 7.7×10^{-8}, whereas higher apparent spontaneous frequencies on SA2 and SA1 reflected limited post-plating division under residual tryptophan. MMS increased revertant frequency above the spontaneous baseline, and the spot assay revealed a clear inhibition zone near the disc, with revertants concentrated outside this region. Spatial analysis indicated that maximal recoverable revertants occurred at intermediate MMS exposure rather than at the highest local concentration. UV produced the highest induced mutation frequency at 20 s, whereas 40 s exposure yielded fewer revertants, consistent with greater lethality at longer exposure. Overall, both MMS and UV were mutagenic under the tested conditions. However, the observed mutation frequency reflected not only mutagenic induction but also cytotoxic loss of viability, and frequencies from different treatments should therefore be compared with caution rather than treated as directly equivalent.

---

2. Introduction

2.1 *E. coli* B strain, tryptophan synthesis, and trp- auxotroph

Escherichia coli B strain WP2 is widely used in mutagenicity assays as it is highly sensitive to a broad range of mutagens and efficiently detects base-substitution events (1). In wild-type *E. coli*, tryptophan biosynthesis is mediated by the trp operon, which contains regulatory elements (*trpR*, promoter, operator, and leader/attenuator) and five structural genes, *trpE*, *trpD*, *trpC*, *trpB*, and *trpA*. These genes encode the enzymes required for the stepwise conversion of chorismate to tryptophan, with *trpE* catalyzing the first committed step of the pathway (2,3).

In the WP2 tester strain, a nonsense mutation in *trpE* disrupts endogenous tryptophan synthesis and produces a trp- auxotrophic phenotype (1). As a result, WP2 trp- cells depend on exogenous tryptophan and cannot grow under tryptophan-deficient conditions. This growth dependence provides a mechanistic basis for detecting revertant mutations in mutagenicity assays.

2.2 Carcinogens, mutagens, and the Ames assay

Many carcinogens act through DNA mutation, which makes mutagenicity testing a central component of chemical safety assessment (4). Mutagens are commonly classified as chemical or physical agents, both of which increase mutation frequency by damaging DNA. Chemical mutagens, such as methyl methanesulphonate (MMS), typically alkylate nucleotide bases, whereas physical mutagens, such as ultraviolet (UV) radiation, generate lesions, including pyrimidine dimers (5,6).

The Ames assay is a bacterial reverse mutation test widely used to assess the mutagenic potential of compounds (7,8). It uses auxotrophic tester strains, including *E. coli* WP2 trp-, in which restoration of gene function is detected as colony formation under selective conditions (1). Because many carcinogens are also mutagenic, the Ames assay provides a rapid and effective screen for potential carcinogens (4,8).

2.3 Spontaneous mutation, induced mutation, and tryptophan reverse mutation

Mutations that restore tryptophan biosynthesis in WP2

trp- cells may arise spontaneously or be induced by external mutagens. Spontaneous mutations occur at low frequency during DNA replication or from endogenous DNA damage (9), whereas induced mutations arise after exposure to agents that increase the rate of DNA base alteration (10).

In the WP2 system, both processes generate trp+ revertants that regain tryptophan independence (1). Reversion may result from base substitutions that restore the original codon in *trpE* (true reversion) (11) or from secondary mutations at other loci, such as Su+ suppressor mutations, which permit translational readthrough of the premature stop codon and restore functional *TrpE* production without correcting the original mutation (12).

Because trp- cells cannot grow without tryptophan, only revertants form colonies under selective conditions. Colony number, therefore, serves as a phenotypic readout of reverse mutation.

2.4 Aim of the study

This study aimed to quantify spontaneous mutation frequency in *E. coli* WP2 trp- and to assess mutation induction by chemical and physical mutagens. Specifically, methyl methanesulphonate (MMS) and ultraviolet (UV) radiation were applied, and mutation frequencies were compared under different experimental conditions using the tryptophan reverse mutation assay. In addition, the study examined qualitative mutation patterns across different exposure conditions.

3. Materials & Method

3.1 Culture preparation

E. coli B strain WP2 trp- cultures were incubated until mid-log phase (OD_{600} = 0.30–0.35, ~2×10^8 cells/ml), then collected by centrifugation at 10,000 rpm for 10 min, and resuspended in PBS to produce a uniform bacterial suspension.

A ten-fold dilution series was prepared to cover dilution levels from 10^0 to 10^{-6}, ensuring recovery of countable plates within the 30–300 colony-forming unit CFU range.

E. coli B strain WP2 trp+ was prepared in parallel as a positive control.

3.2 Agar preparation

Nutrient agar (NA), which contains non-limiting tryptophan, supported the normal growth of *E. coli* B strain WP2 trp-. Selective agar (SA) contained limited or no tryptophan and was used to permit limited post-plating division, allowing revertants to segregate from parental cells. Three selective media were used: SA3 (0 µg/ml tryptophan), SA2 (1 µg/ml tryptophan), and SA1 (0.25 µg/ml tryptophan).

3.3 Plating design

Bacterial suspensions and mutagen treatments were applied to Plates 1–12 (Table 1).

Plate 1 carried trp+, and trp- strains streaked separately. Plates 2–6 received 0.1 ml of undiluted or diluted WP2 trp- suspension, and were spread evenly across. For the MMS spot assay (Plates 7–9), a sterile filter disc was placed at the center of each plate. Discs were loaded with 25 µL of 1% MMS (Plate 7), 2% MMS (Plate 8), or sterile water (Plate 9). For MMS liquid treatment (Plate 10), 0.5 ml of culture was incubated with 5 µL of 1% MMS at 37°C for 30 min, and 0.1 ml of the treated suspension was then plated on SA1. For UV mutagenesis (Plates 11–12), 0.1 ml of undiluted culture was spread on SA2, and irradiated without lids at 10 J/m² UV for 20 s (Plate 11) or 40 s (Plate 12). All plates were incubated at 37°C overnight before colony counting.

Plate number	*Culture dilution*	*Plate type*	*Treatment*
1	10^{0}	*SA3*	*Trp+ strain & Trp- strain*
2	10^{0}	*SA3*	*None*
3	10^{0}	*SA2*	*None*
4	10^{0}	*SA1*	*None*
5	10^{-5}	*NA*	*None*
6	10^{-6}	*NA*	*None*
7	10^{0}	*SA2*	*Spot test with 1 % MMS*
8	10^{0}	*SA2*	*Spot test with 2 % MMS*
9	10^{0}	*SA2*	*Spot test control (water)*
10	10^{0}	*SA1*	*Treat with 1 % MMS, then plate*
11	10^{0}	*SA2*	*Irradiate with UV light for 20 s*
12	10^{0}	*SA2*	*Irradiate with UV light for 40 s*

Table 1. *Experimental design of Plates 1–12, showing culture dilution, plate type, and treatment for each plate.*

3.4 Control design

Negative, positive, procedural, and viability controls were included to validate the assay and separate induced mutation from background reversion.

Plate 1 served as a positive and phenotype control. Normal growth of trp+ and only minimal growth of trp- verified both the auxotrophic defect for tryptophan-free medium survival, and the absence of tryptophan contamination in SA3.

Plates 2–4 served as negative controls for spontaneous reversion for estimating true and apparent spontaneous mutation frequencies under different tryptophan conditions. The expected background reversion rate was low, and the apparent frequency was expected to increase with tryptophan concentration because residual growth permits limited post-plating division and segregation of newly formed revertants. These values were subsequently used as plate-specific spontaneous baselines in induced mutation analysis.

Plates 5 and 6 served as viability controls. Diluted cells were plated on NA to estimate the number of viable cells in the original suspension, providing the denominator for mutation frequency calculations. This also ensured recovery of countable plates within the accepted 30–300 CFU range.

Plate 9 served as the procedural control for the MMS spot assay. Sterile water was applied as a control for disc placement, liquid diffusion, and local handling effects independent of mutagen exposure.

Together, these controls established the appropriate baseline for each condition, verified assay selectivity, and ensured that each comparison differed by only one experimental variable.

3.5 Colony counting

Colonies were counted by different methods according to plate type and colony density.

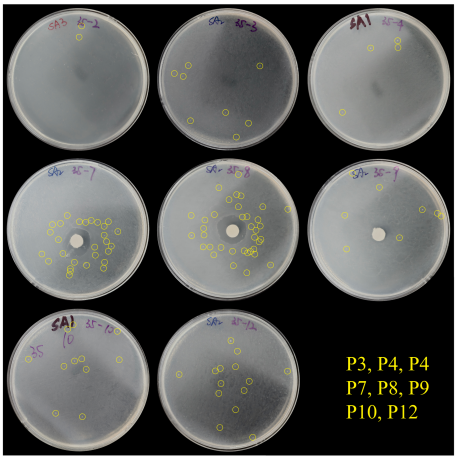
Plate 1 served as a qualitative control and was not included in numerical counting. Plates 2–4, 7–10, and 12 contained clearly resolved colonies in countable numbers, so colonies were counted manually (Figure 1A). Final counts for these plates were obtained as the mean of counts recorded independently by three experimenters. Plates 5 and 6 contained larger numbers of colonies and were therefore counted using ImageJ (version 1.54p), followed by manual inspection and adjustment where necessary to obtain the final count (Figure 1B).

Plate 11 could not be counted reliably by either direct manual counting or ImageJ-assisted analysis because

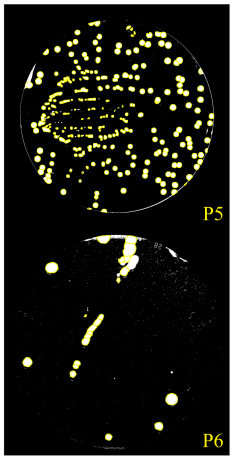
colonies were both numerous and tiny. Colony distribution was also uneven, with one sparse half and one dense half. Consequently, the sparse region was counted directly as N_{sparse}. The dense region, defined as one-half of the plate with radius r_{plate}, was estimated by subsampling. Three non-overlapping circles of radius r_{circle} were placed at randomly generated coordinates within the dense half, and colonies within each circle ($N_{circle,\,i}$) were counted. The mean count from these three circles was then scaled to the area of the dense half to estimate N_{dense}. Total colony number for Plate 11 (N_{total}) was then calculated using equations (1) & (2) (Figure 1C).

$$N_{total} = N_{sparse} + N_{dense} \quad (1)$$

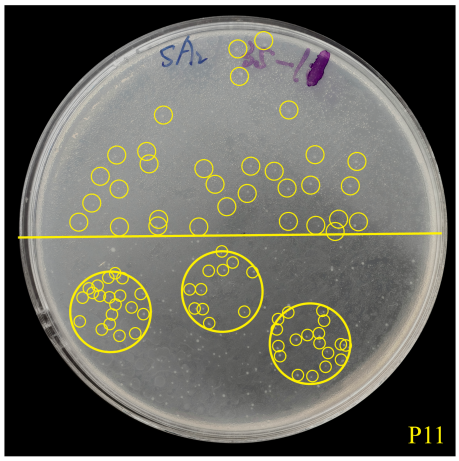
$$N_{dense} = \frac{N_{circle,1} + N_{circle,2} + N_{circle,3}}{3} \times \frac{r_{plate}^2}{2r_{circle}^2} \quad (2)$$



(A)



(B)



(C)

Figure 1. *Colony counting workflow.* ***(A)*** *Plate 2-4, 7-10, 12: manually counted. Each yellow circle indicates one colony.* ***(B)*** *Plate 5-6: ImageJ-assisted counting. Bright objects correspond to colonies detected.* ***(C)*** *Plate 11: subsampling-based estimation. Estimation was based on equation (1) & (2)*

4. Results & Analysis

4.1 Colony counts and plate description

Twelve plates were analyzed under control and mutagen treatment conditions. Raw colony counts are summarized in Table 2, Figure 2A, and raw plate images are shown in Figure 1 and Figure 4 in the Appendix.

On Plate 1, the trp+ side showed clear growth along the streak, whereas the trp- side showed only a faint streak with little or no visible colony formation. This confirmed the expected trp- auxotrophic phenotype under tryptophan-free conditions.

Plates 2–4 produced only small numbers of colonies: 2 on SA3 (Plate 2), 8 on SA2 (Plate 3), and 4 on SA1 (Plate 4). These low counts are consistent with the low background frequency of spontaneous reversion. The higher counts on SA2 and SA1 relative to SA3 also indicate that limited residual tryptophan permitted a small amount of post-plating division, allowing additional revertants to segregate from parental cells.

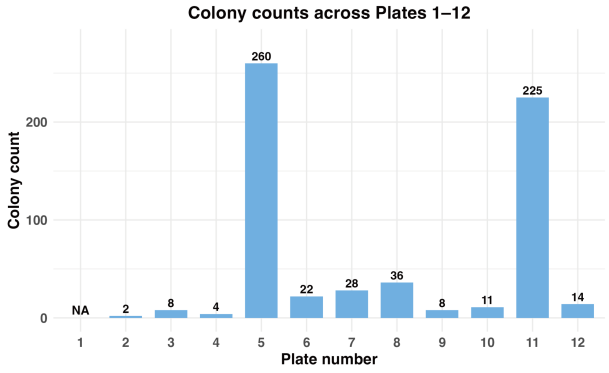
On nutrient agar, Plate 5 (10^{-5} dilution) and Plate 6 (10^{-6} dilution) produced substantially more colonies. Plate 5 yielded 260 colonies, which fell within the accepted countable range of 30–300 CFU, whereas Plate 6 yielded 22 colonies and was therefore used only as a supporting check. Colonies on NA were also larger and less transparent than those on selective agar, consistent with the background that tryptophan biosynthesis is an energy and material-consuming process for bacteria (13).

For the MMS spot assay, Plate 7 (1% MMS) and Plate 8 (2% MMS) both showed a central filter disc surrounded by a visible inhibition zone, with scattered colonies outside this region. Final colony counts were 28 and 35, respectively. In contrast, Plate 9 (water control) showed no obvious inhibition zone and only 8 colonies. Plate 10, which received 1% MMS by liquid treatment rather than disc diffusion, produced 11 discrete colonies distributed across the plate.

Plate 11 (UV, 20 s) showed a heterogeneous pattern, with one sparse region and one densely populated region of small colonies. The total colony number was therefore estimated by subsampling and gave a final count of 225. By contrast, Plate 12 (UV, 40 s) showed only 14 discrete colonies.

Plate number	***Colony count***	***Plate number***	***Colony count***
1	*NA*	*7*	*28*
2	*2*	*8*	*36*
3	*8*	*9*	*8*
4	*4*	*10*	*11*
5	*260*	*11*	*225*
6	*22*	*12*	*14*

Table 2. *Colony counts for Plates 1–12 (Plate 1 was qualitative and therefore treated as uncountable).*



(A)

Figure 2. (A) *Colony counts distribution for Plates 1–12 (Plate 1 was qualitative and therefore treated as uncountable).*

4.2 Viable cell count (CFU) estimation

Plate 5 (10^{-5} dilution) yielded 260 colonies within range of 30-300 CFU, and was therefore used to estimate viable cell concentration using equation (3), where N is the counted number of colonies, D is the dilution factor, and V is the plated volume (ml).

$$CFU/ml = \frac{N}{D \times V} \quad (3)$$

Substitution of the Plate 5 values gave a viable cell concentration of 2.6×10^{8} CFU/ml. Plate 6 yielded 22 colonies, close to the expected tenfold reduction from Plate 5, which supports both the CFU estimate and the correctness of the serial dilution. The calculated value was also close to the expected concentration of approximately 2×10^{8} cells/ml. This value was used as the denominator in all subsequent mutation frequency calculations.

4.3 Spontaneous mutation frequency

Spontaneous mutation frequency was calculated using equation (4), where f is the spontaneous mutation frequency, $N_{revertants}$ is the number of revertant colonies, and V_{plate} is the volume of undiluted culture plated (ml).

$$f = \frac{N_{revertants}}{(CFU/ml) \times V_{plate}} \quad (4)$$

The true spontaneous mutation frequency was estimated from Plate 2 (SA3), which contained no tryptophan and therefore did not permit post-plating division or segregation of newly formed revertants. Using the Plate 2 count, the true spontaneous mutation frequency was estimated as 7.7×10^{-8} (Table 3).

Apparent spontaneous mutation frequencies were then calculated for SA2 and SA1 to provide medium-matched baselines for later induced mutation analysis. Plate 3 (SA2) gave an apparent spontaneous mutation frequency of 3.1×10^{-7}, whereas Plate 4 (SA1) gave 1.5×10^{-7}. Both values exceeded the SA3 estimate, consistent with limited post-plating division in the presence of residual tryptophan. For subsequent mutagenesis assays, induced mutation frequency was calculated using equation (5) to remove the contribution of spontaneous background under the corresponding selective condition.

$$f_{induced}^{(SAi)} = f_{induced,apparent}^{(SAi)} - f_{apparent}^{(SAi)} \quad (5)$$

4.4 MMS-induced mutation analysis

Plate 9 showed no inhibition zone and only sparse, randomly distributed colonies, with colony count identical to the SA2 baseline in Plate 3, indicating that neither disc placement nor liquid diffusion alone generated the observed pattern. This supported interpretation of the MMS spot plates as mutagen-

specific rather than procedure-dependent.

Mutation frequencies for MMS-treated plates were first calculated for the spot assay (Plates 7–8) and liquid treatment assay (Plate 10) (Table 3).

Spatial pattern analysis was then conducted for spot assay in Plates 7 and 8 using raw colony counts. The disc radius (*r*) was used as the unit distance, and concentric circular zones with radius *ir (i = 1…10)* were defined from the disc center (Figure 3A). Colonies within each zone were counted and further analyzed as newly observed and cumulative counts (Figure 3B, C). No colonies were detected within the disc region (0–1) or the immediately adjacent region (1–2), corresponding to the inhibition zone, indicating strong lethality of high local MMS concentrations to WP2 trp- cells. Colonies first appeared in the 2–3 interval and increased sharply thereafter. Newly observed colonies peaked at intermediate distances, approximately 3–5 disc-width units, in both plates, then declined towards zero at larger distances. The cumulative curves rose most rapidly between 3 and 6 disc-width units before reaching a plateau. This pattern suggests that maximal recoverable revertants occurred where mutagenic induction was sufficient to generate reversion but local cytotoxicity had declined enough to permit survival. Plate 8 (2% MMS) reached a higher final cumulative count than Plate 7 (1% MMS), with 36 and 28 colonies, respectively, indicating a greater overall revertant recovery rate.

Plate 10, in contrast, involved uniform liquid exposure rather than disc diffusion and therefore showed no inhibition zone. The increased revertant frequency on this plate compared to the SA1 baseline indicates that mutagenesis was attributable to MMS itself rather than to spatial concentration gradients generated in the spot assay.

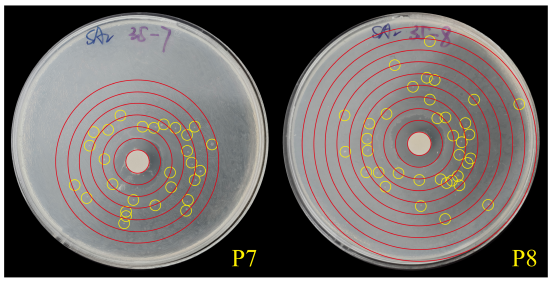
4.5 UV-induced mutation analysis

Mutation frequencies for UV-treated plates (Plates 11–12) were also calculated for reference (Table 3).

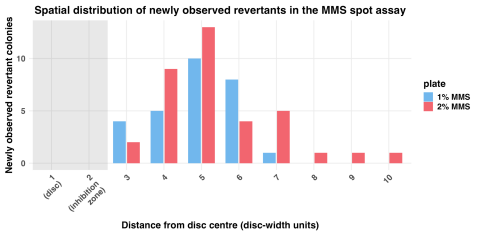
UV exposure for 20 s produced the highest mutation frequency among all induced treatments in this study, whereas 40 s exposure yielded fewer colonies, consistent with greater cytotoxicity at longer exposure. Because only two exposure times were tested, the UV response could not be resolved further, and more detailed interpretation remains limited.

Plate number	*Apparent mutation frequency*	*True mutation frequency*
2	*NA*	7.70×10^{-8}
3	3.10×10^{-7}	*NA*
4	1.50×10^{-7}	*NA*
7	1.08×10^{-6}	7.69×10^{-7}
8	1.38×10^{-6}	1.08×10^{-6}
9	3.10×10^{-7}	*0*
10	4.23×10^{-7}	2.69×10^{-7}
11	8.65×10^{-6}	8.35×10^{-6}
12	5.38×10^{-7}	2.31×10^{-7}

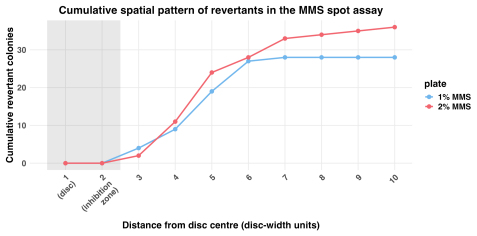
Table 3. *Apparent and corrected mutation frequencies for spontaneous and mutagen-treated plates. Corrected mutation frequencies were obtained using equation (5).*



(A)



(B)



(C)

Figure 3. *Spatial analysis of revertant distribution in the MMS spot assay.* ***(A)*** *Plates 7 and 8 are annotated with concentric distance zones centered on the disc. The disc radius (r) was used as the unit distance.* ***(B)*** *Number of newly observed revertant colonies within each distance interval from the disc center for MMS treatments.* ***(C)*** *Cumulative revertant colony counts as a function of distance from the disc center for MMS treatments.*

5. Conclusion & Discussion

5.1 Conclusion

This study quantified spontaneous and induced mutation frequencies in *E. coli* WP2 trp- under controlled exposure to the chemical mutagen MMS and the physical mutagen UV.

The true spontaneous mutation frequency was estimated as 7.7×10^{-8}, estimated using culture with approximately 2.6×10^{8} CFU/ml, whereas the apparent spontaneous frequencies on SA2 and SA1 were 3.1×10^{-7} and 1.5×10^{-7}, respectively. These higher values reflect additional cell division and segregation of revertants under tryptophan-limited conditions.

MMS increased the mutation frequency above the spontaneous background, and 2% MMS produced more revertants than 1% MMS. Spatial analysis of the spot assay showed a clear inhibition zone near the disc and a peak in revertant recovery at intermediate distances, indicating that maximal observable reversion occurred where mutagenic induction remained effective but cytotoxicity had declined sufficiently to permit survival.

UV produced the highest induced mutation frequency at 20 s, whereas 40 s exposure reduced the number of detectable revertants, consistent with increased lethality at longer exposure.

Overall, both MMS and UV were mutagenic under the tested conditions. However, the observed mutation

frequency reflected not only mutagenic induction but also cytotoxic loss of viability. Because the two mutagens differed in exposure mode and dose structure, their frequencies are informative but not strictly directly equivalent for comparison.

5.2 Discussion

5.2.1 Control settings

The control design shaped not only assay validation but also result interpretation. The contrast between SA3 and SA1/SA2 confirmed the trp- phenotype, but more importantly, established that the observed colony number depends on both pre-existing revertants and the extent of post-plating division permitted by the medium. The control plates, therefore, provided the necessary calibration for interpreting apparent and induced mutation frequencies under different selective conditions.

5.2.2 MMS induced mutation

The spot and liquid treatment assays resolved different aspects of MMS action. The spot assay revealed the spatial consequences of a diffusion-based concentration gradient. In contrast, the liquid treatment assay confirmed that the observed increase in revertants was attributable to MMS itself rather than to local diffusion effects.

The spot assay further showed that mutagenicity and cytotoxicity were competing outcomes rather than independent effects. The inhibition zone around the disc indicates that high local MMS concentrations were strongly lethal to WP2 trp- cells. In contrast, the greater number of colonies at intermediate distances shows that maximal revertant recovery did not occur at the highest dose. Instead, recoverable revertants peaked where mutagenic induction remained sufficient to generate reversion, but cytotoxicity had declined enough to permit survival. The absence of a simple monotonic relationship between local MMS concentration and colony recovery is consistent with previous reports (14).

5.2.3 UV light induced mutation

The UV results showed a similar mutagenicity–cytotoxicity trade-off. Recoverable mutation frequency was much higher after 20 s exposure than after 40 s, although the longer exposure still produced induced revertants. This pattern is broadly consistent with the literature (15), which reported a non-monotonic, three-phase relationship between UV intensity and bacterial mutation frequency rather than a simple linear increase. Although that study used UV intensity rather than exposure time, the two variables are physically related through dose and therefore remain qualitatively comparable.

5.2.4 Limitations

Several limitations, beyond the lack of biological replicates or batch-effect removal, constrain the interpretation of these results.

First, both MMS and UV produced substantial cytotoxicity, which means that some revertants may have formed but failed to survive long enough to be detected as colonies. This can lead to the potential underestimation of the true spontaneous mutation frequency.

A further limitation is that exposure modes differed across assays. MMS spot treatment, MMS liquid treatment, and UV irradiation can each yield a mutation frequency, but they are not dose-equivalent systems and cannot be converted into directly comparable exposure measures. The comparison is therefore biologically informative but not strictly quantitative. Dose sampling was also limited: the MMS spot assay contained only two nominal concentrations, each of which generated an unquantified diffusion gradient, while the UV experiment included only two exposure times, sufficient to show trend direction but insufficient to resolve the full response curve or define an optimum precisely.

These limitations do not overturn our main findings, but they do define the interpretation scope. The present data support a relative interpretation of

mutagenic patterns under the tested conditions rather than strict quantitative equivalence across different mutagen treatments.

5.2.5 Future directions

A promising direction arises from the genetic basis of trp+ reversion. As noted in Introduction 2.3, trp- reversion in the WP2 system can occur through two distinct mechanisms: true reversion of the original *trpE* nonsense mutation or secondary suppressor mutation at another locus. These cannot be distinguished by colonies in the present experiment. This ambiguity could be resolved in future work by sequencing the *trpE* locus, where restoration of the original codon would indicate true reversion. In contrast, retention of the nonsense mutation would suggest suppression. Alternatively, genetic transfer of the *trpE* region could determine whether the causative mutation resides within *trpE* or at secondary loci. Functional assays assessing the ability of revertants to suppress other nonsense mutations would further help identify tRNA-mediated suppressor effects.

Second, assay coverage should be expanded beyond the current WP2 system. Because WP2 primarily detects base-substitution events and lacks metabolic activation, mutagens that require bioactivation or induce other classes of DNA damage may remain undetected. Inclusion of an S9 metabolic activation system (16,17), additional tester strains, or complementary assays would therefore provide a more comprehensive assessment of mutagenic potential, like the example we mentioned in Question 7.

6. Answers to Questions

1. Colony counts for Plates 1–12 are summarized in Table 2, Figure 2A, with detailed images shown in Figure 1.

2. From Plate 5, the undiluted culture was estimated to contain 2.6×10^{8} CFU/ml. Therefore, Plates 1–4 and 7–12, each receiving 0.1 ml undiluted culture, contained 2.6×10^{8} cells per plate. Plate 5, which received 0.1 ml of a 10^{-5} dilution, contained 2.6×10^{2} cells, and Plate 6, which received 0.1 ml of a 10^{-6} dilution, contained 260 cells.
Spontaneous mutation frequency was calculated using equation (4) discussed in Result 4.3. Based on Plate 2 (SA3), the true spontaneous mutation frequency was 7.7×10^{-8}.

3. The apparent spontaneous frequencies on SA2 (Plate 3) and SA1 (Plate 4) were 3.1×10^{7} and 1.5×10^{7}, respectively, as discussed in Result 4.3.
These are only apparent frequencies because SA2 and SA1 contain residual tryptophan, which permits limited post-plating division of trp- cells before tryptophan is exhausted. Additional revertants can therefore arise and segregate during this brief growth period. By contrast, SA3 contains no tryptophan, so only revertants already present at plating can form colonies. The SA2 value exceeds the SA1 value because SA2 contains more tryptophan and thus permits more residual growth, increasing the opportunity for new revertants to appear.

4. As shown in Result 4.4 and Table 3, MMS-induced mutation frequencies were first calculated as apparent frequencies and then corrected by subtracting the corresponding spontaneous background under the same selective condition (SA). All apparent and corrected MMS mutation frequencies are listed in detail in Table 3.

5. All apparent and corrected UV light induced mutation frequencies are listed in detail in Table 3.
Corrected mutation frequencies are 8.35×10^{-6} for Plate 11 (20 s), and 2.31×10^{-7} for Plate 12 (40 s).
Among all induced treatments, UV at 20 s gave the highest induced mutation frequency.
Direct comparison with MMS is not valid, because MMS and UV act through different mechanisms, were applied by different procedures, and were assessed under different spatial and recovery conditions.

6. The MMS spot plates showed a clear inhibition zone around the disc. Colonies appeared outside it.

The water control showed only sparse and evenly distributed colonies.
These plates are not suitable for precise counting, because MMS is not uniform across the plate and varies continuously with diffusion distance from the disc. Colony number reflects both mutagenicity and cytotoxicity along a gradient rather than a single defined dose.
The 2% MMS plate gave more colonies than 1% (36 vs. 28). So overall revertant recovery was higher. But this is not a true dose–response result. Only a rough comparison.

7. Not sure, at least cannot conclude compound Z as non-mutagen.
The reported carcinogenicity in rodents suggests that Z may act through mechanisms. The WP2 system mainly detects base-substitution mutations and lacks metabolic activation. So, some mutagens may be missed, especially promutagens, or compounds causing other classes of DNA damage.
To test mutagenicity more reliably, the experiment should be redesigned to include an S9 metabolic activation system (16,17) or additional tester strains.

8.
1). Plating more cells (avoid over-crowding) to improve the probability of detecting revertants.
2). Refining the exposure range through additional pilot experiments, so that treatment conditions fall within an informative mutagenic window rather than being too weak or excessively lethal.
3). Optimizing incubation conditions, including longer incubation where appropriate, to allow slower-growing revertants to develop into visible colonies.
4). For compounds that require metabolic activation (e.g.: compound Z in Question 7), inclusion of an S9 system would further increase detection sensitivity (16,17).

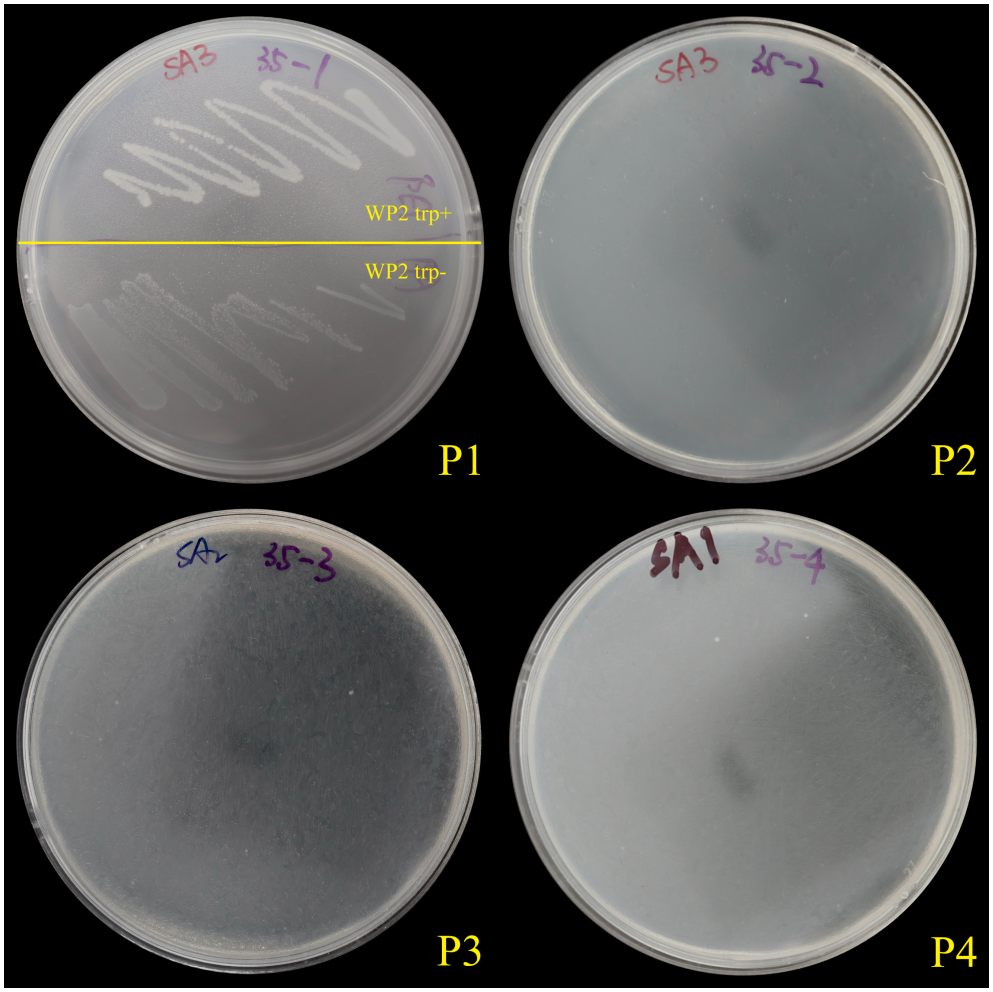
7. Reference

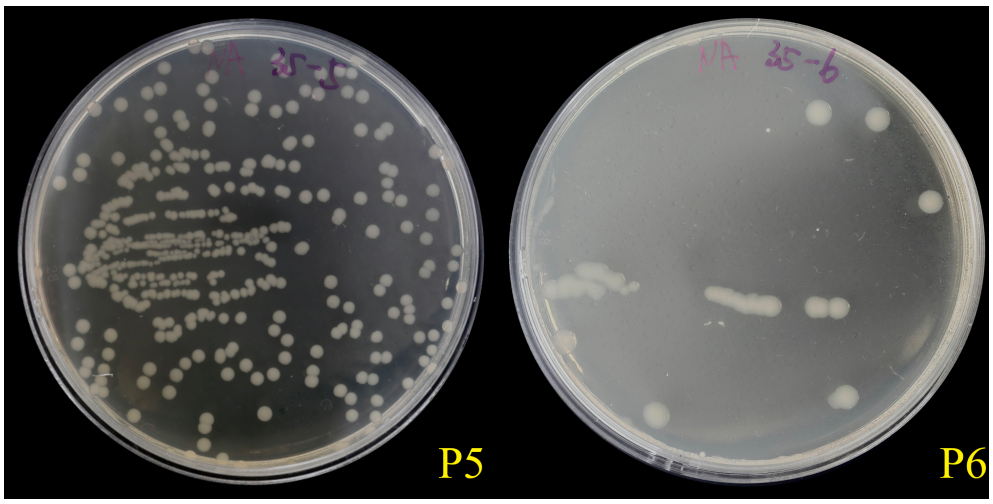
1. Mortelmans, K. and Riccio, E.S. (2000) The bacterial tryptophan reverse mutation assay with Escherichia coli WP2. Mutation Research - Fundamental and Molecular Mechanisms of Mutagenesis, 455, 61-69.
2. Martins, N.F., Viana, M.J.A. and Maigret, B. (2024) Fungi Tryptophan Synthases: What Is the Role of the Linker Connecting the α and β Structural Domains in Hemileia vastatrix TRPS? A Molecular Dynamics Investigation. Molecules, 29.
3. Yanofsky, C. (1981) Attenuation in the control of expression of bacterial operons. Nature, 289, 751-758.
4. Ames, B.N., McCann, J. and Yamasaki, E. (1975) Methods for detecting carcinogens and mutagens with the Salmonella/mammalian-microsome mutagenicity test. Mutat Res, 31, 347-364.
5. Beranek, D.T. (1990) Distribution of methyl and ethyl adducts following alkylation with monofunctional alkylating agents. Mutat Res, 231, 11-30.
6. Cadet, J., Douki, T. and Ravanat, J.L. (2015) Oxidatively generated damage to cellular DNA by UVB and UVA radiation. Photochem Photobiol, 91, 140-155.
7. Zeiger, E. (2019) The test that changed the world: The Ames test and the regulation of chemicals. Mutat Res Genet Toxicol Environ Mutagen, 841, 43-48.
8. Vijay, U., Gupta, S., Mathur, P., Suravajhala, P. and Bhatnagar, P. (2018) Microbial Mutagenicity Assay: Ames Test. Bio Protoc, 8, e2763.
9. Lind, P.A. and Andersson, D.I. (2008) Whole-genome mutational biases in bacteria. Proc Natl Acad Sci U S A, 105, 17878-17883.
10. Friedberg, E.C. (2006) DNA Repair and Mutagenesis. ASM Press.
11. Ohta, T., Tokishita, S., Tsunoi, R., Ohmae, S. and Yamagata, H. (2002) Characterization of Trp(+) reversions in Escherichia coli strain WP2uvrA. Mutagenesis, 17, 313-316.
12. Sikora, A., Grzesiuk, E. and Janion, C. (2012) In Mishra, R. (ed.), Mutagenesis. IntechOpen, London.

13. Priya, V.K., Sarkar, S. and Sinha, S. (2014) Evolution of tryptophan biosynthetic pathway in microbial genomes: a comparative genetic study. Syst Synth Biol, 8, 59-72.
14. Smith, C. and Perfetti, T. (2020) Statistical treatment of cytotoxicity in Ames bacterial reverse mutation assays can provide additional structure–activity relationship information. Toxicology Research and Application, 4, 239784732091163.
15. Doudney, C.O. (1976) Complexity of the ultraviolet mutation frequency response curve in Escherichia coli B/r: SOS induction, one-lesion and two-lesion mutagenesis. J Bacteriol, 128, 815-826.
16. Hakura, A., Suzuki, S. and Satoh, T. (1999) Advantage of the use of human liver S9 in the Ames test. Mutation Research/Genetic Toxicology and Environmental Mutagenesis, 438, 29-36.
17. Thomas, D.N., Wills, J.W., Tracey, H., Baldwin, S.J., Burman, M., Williams, A.N., Harte, D.S.G., Buckley, R.A. and Lynch, A.M. (2024) Ames test study designs for nitrosamine mutagenicity testing: qualitative and quantitative analysis of key assay parameters. Mutagenesis, 39, 78-95.

8. Appendix

All raw photos for Plates 1-12 are listed here for reference.





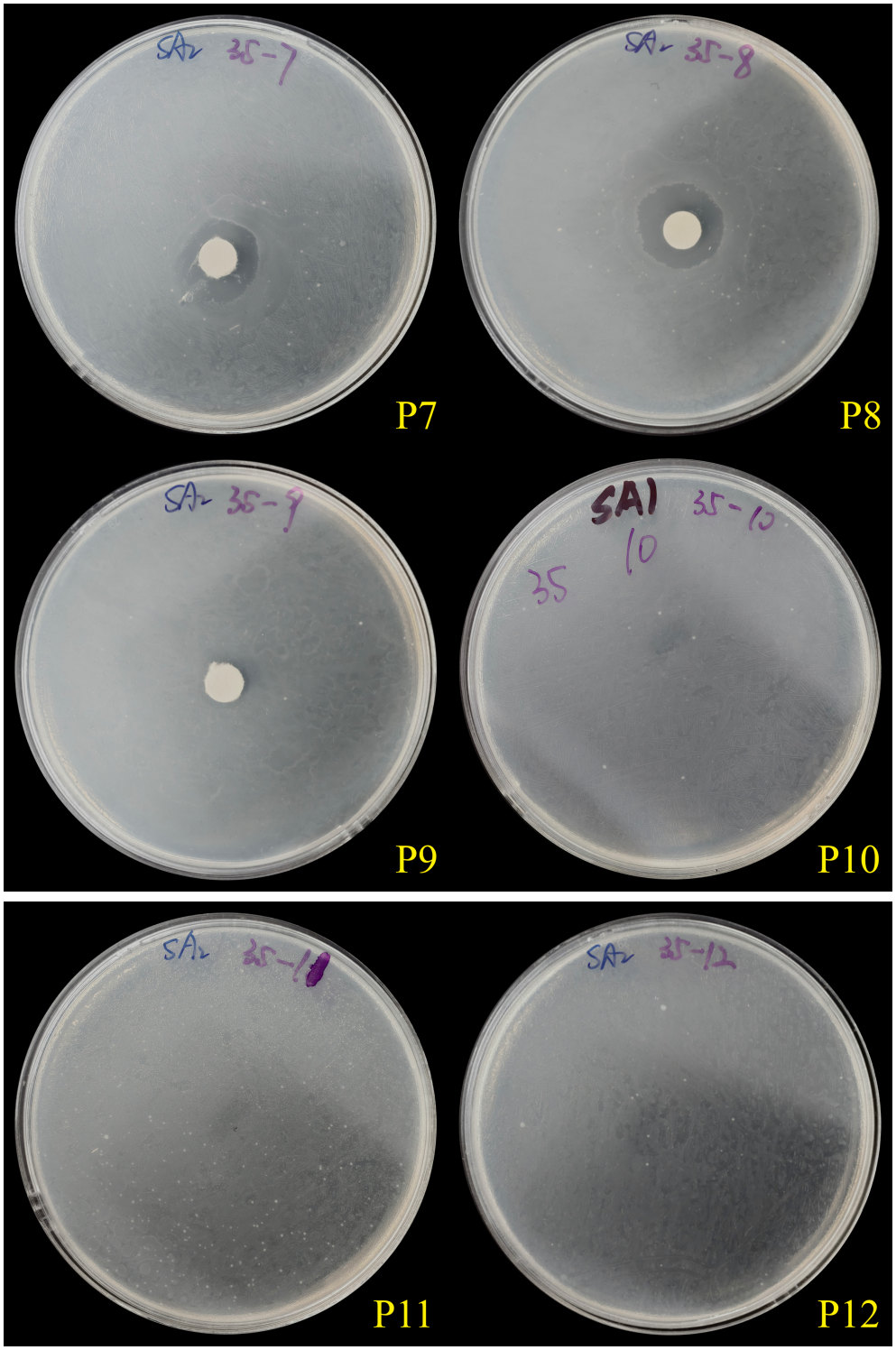


Figure 4. *Raw photos for Plates 1-12.*